

Descriptive Statistics

Easy Level

Q1. Understanding Central Tendency

A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12]. What is the most representative value of their weekly sales, and why?

Ans. The sales are : 10,11,12,12,13,14,15(this specified to smallest to largest.)

Mean :- $(10 + 12 + 11 + 15 + 14 + 13 + 12) / 7 = 12.43$

Median :- The middle value in the sorted list = 12

Mode :- The number that appears most often = 12

The Ans 12 is represent both of the median and mode.

Why??

The median is 12 : When line the numbers up

(10,11,12,12,13,14,15), 12 exactly in the middle.

The mode is 12 : It is the only numbers that repeats, making is the most typical day.

The mean (12.43) is impractical : Can't easily track 0.43 of a dozen in daily sales, 12 is a real, recurring data point.

Q2. Mean in Real Life

A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19].

What is the mean score, and what does it tell us about the class's performance?

Ans. The mean score is 16.28

Add all the scores together $(12 + 15 + 14 + 16 + 18 + 20 + 19 = 114)$

Then divide by 7 students giving you $114 / 7 = 16.28$

This means the average student scored about 16 out of 20, which is pretty good it tells the teacher that the class as a whole, understood the material well most students performed near or above this mark suggesting the lesson was effective.

Q3. Mode is Real Life

A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9]. What is the mode, and why is this information useful for the store manager?

Ans. The mode is 8

The store's manager now knows that size 8 sells the most. This is useful because it helps the manager order more of size 8 to keep enough stock on the shelf. It prevents running out of the most demanded size and avoids losing customers.

Medium Level

Q4. Median in Real Life

A car dealer notes the prices of used cars: [\$8,000, \$9,500, \$10,200, \$11,000, \$50,000]. Why is the median a better measure than the mean in this case? Calculate the median.

Ans. The median is \$10,200

Why the median is better??

The \$50,000 car is way higher than the rest; it is an outlier. If you calculate the mean, it comes out to about \$17,740, which doesn't really represent what most cars cost on that lot, but the median ignores that extreme value and gives you a truer picture of the typical car price. The most cars here cost around \$10,200.

Q5. Dispersion Introduction

A student times how long it takes to finish a puzzle each day: [25, 30, 27, 35, 40]. What does the range tell us about the variation in the student's puzzle-solving time?

Ans. The range is 15 mins.

To find it: $40 - 25 = 15$

The range shows that the student's puzzle-solving time varied by 15 mins across the days. On the fastest day it took 25 mins and on the slowest it took 40 mins.

Q6. Range in Action

A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120]. Find the range. How can this help the farmer in planning his package?

Ans. The range is 22kg.

To find it: $120 - 98 = 22$

The farmer's weekly harvest varied by 22kg; the lowest was 98kg and the highest was 120kg.

This helps the farmer plan packaging smartly. Since the harvest varies by 22kg, the farmer should prepare enough packaging materials to handle up to 120kg. Knowing the range means no

surprise, better preparation and less waste in packaging and planning.

Q7. Variance for Decision-Making (Medium)

Two delivery companies track delivery delays (in minutes).

Company A: variance = 6

Company B: variance = 15

Which company is more consistent, and why?

Ans. Company “A” is more consistent. Because variance measures how spread out the delivery times are from the average; the lower the variance, the more consistent the performance.

Company A has a variance of 6; their delivery delays are close together and predictable. Company B has a variance of 15; their delays jump around a lot and are harder to predict.

So for a business or customer who values punctuality and consistency, company A is the better choice.

Hard Level

Q8. Standard Deviation in Context

A finance student compares the daily price fluctuations of two cryptocurrencies.

Coin A: standard deviation = \$30

Coin B: standard deviation = \$120

Which coin is riskier to invest in, and why?

Ans. Coin “B” is the riskier because standard deviation shows how much the price moves up and down from the average. Coin B's deviation of \$120 means wild, unpredictable price changes, while coin A's \$30 means it stays relatively calm and stable.

Coin B is a rollercoaster, Coin A is a smooth road. Higher risk, higher reward, but also higher chances of loss.

Q9. Combining Measures

A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410] Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Ans. Add all values: $400+420+390+450+410=2070/5=414$

The ans is mean = 414 kWh

Standard deviation = 20.7 kWh

The values don't stray far from 414, giving a small SD, The family uses an average of 414 kWh per month, and their usage only varies by about 20.7 kWh

Q10. Practical Application

A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21]. Find the mean, median, mode, range, and standard deviation. What insights can these measures provide about the player's scoring performance?

Ans. Step 1: Mean = 19.625

Add all: $15+18+20+22+25+17+19+21 = 157 \div 8 = 19.625$

Step 2: Median = 19.5

Arrange in order: [15, 17, 18, 19, 20, 21, 22, 25] Middle two values: $(19+20) \div 2 = 19.5$

Step 3: Mode = None

Every score appears only **once** — no mode.

Step 4: Range = 10

$25 - 15 = 10$

Step 5: Standard Deviation ≈ 2.95

Values stay close to the mean, giving a small SD of about **2.95**

DRIVE LINK: <https://drive.google.com/drive/folders/1ZlFjacMcn7A-Gi0Qmzbq18jmXQvYluqH?usp=sharing>